

DESIGN OF DIGITAL CLOCK USING BASIC LOGIC GATES AND AN OSCILLATING CIRCUIT

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AIM:

To design a digital clock using basic logic gates and an oscillator and to simulate its functionality and develop a PCB for prototyping.

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COMPONENTS REQUIRED:

1. Logic Gates: AND, OR, NOT, NAND, NOR, XOR, XNOR (as per design requirements).
2. Oscillator Circuit: Crystal Oscillator (eg:- 1MHz or 32.768 kHz) or a 555 Timer IC in a stable mode.
3. Counters: 7490 (Decade Counter), 7493 (4-Bit Binary Counter).
4. Decoders/Drivers: 7447 (BCD to 7-segment decoder).
5. Display: 7-segment displays (Common cathode/anode).
6. Resistors and Capacitors: Various values for timing and logic level adjustments.
7. Power Supply: +5V DC.
8. Breadboard and Connecting Wires.



BLOCK DIAGRAM:

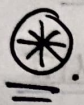
1. Oscillator → Generates clock pulses (eg:- 1 Hz) from a crystal oscillator.
2. Frequency Divider → Converts high-frequency pulses into 1 Hz pulses.
3. Seconds Counter → Counts from 00 to 59.
4. Minutes Counter → Counts from 00 to 59.
5. Hours Counter → Counts from 00 to 12 or 00 to 24.
6. Decoder and Display → Converts BCD output to 7-segment display.



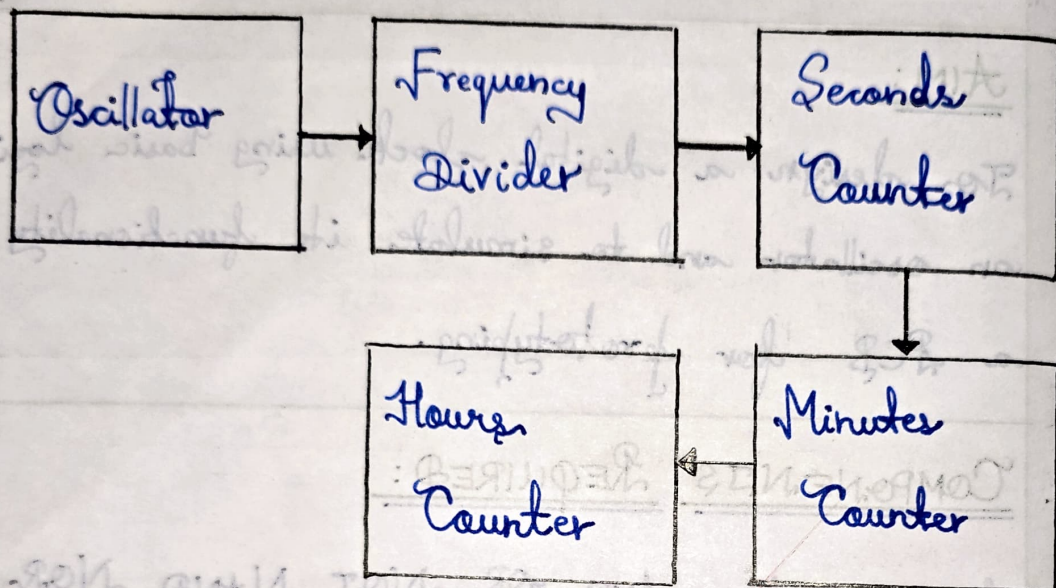
CIRCUIT DESIGN:

Step 1: Clock Pulse Generation:-

- Use a 555 Timer in a stable mode to generate a 1 Hz signal.
- Alternatively, use a crystal oscillator with a frequency divider (eg:- 7490 decade counter) to generate 1 Hz pulses.



BLOCK DIAGRAM:





Step 2: Counter Design:-

- Use 7490 decade counters for seconds and minutes.
- First 7490 counts 0-9 (units place).
- Second 7490 counts 0-5 (tens place).
- Use 7492 or 7493 binary counters for hours (up to 12 or 24).
- Reset logic is implemented using NAND gates to reset counters at the appropriate count.

Step 3: Display Circuit:-

- Connect the BCD outputs of counters to 7447 BCD-to-7-segment decoders.
- The output of the decoder is connected to 7-segment displays for visualization.

Step 4: Logic Implementation:-

- Logic gates (AND, OR, NAND) are used to reset the counters at specific values (eg:- resetting seconds at 60, hours at 12 or 24).



PROCEDURE:

1. Build the clock pulse generator using a 555 timer or crystal oscillator.
2. Connect the pulse generator output to the first counter (seconds counter).

3. Connect cascading counters for minutes and hours.
4. Implement reset logic using NAND gates to ensure proper counting.
5. Connect BCD to 7-segment decoders to drive the display.
6. Power the circuit and observe the counting operation.
7. Verify reset conditions and adjust if needed.



OBSERVATIONS:

- Verify that seconds count from 00 to 59.
- Ensure minutes and hours increment correctly.
- Check if the reset logic works properly at specific counts.



RESULT:

Thus, the digital clock using basic logic gates and an oscillator was designed and simulated and a PCB prototyping for the digital clock circuit was developed. The clock counts time in hours, minutes and seconds, using counters and displays the output on 7-segment displays.

Connect various counters for minor and major

count of shop GMA's give signal that the counter
proper working.

Connect B.C. to F-2000 counter to show the
display.

From the circuit above the counter information

is sent to the counter and the counter is

Corrected
2/24/66

OBSERVATIONS:

Verify that counter count from 0 to 25.

Count minutes and hours and minutes correctly

Check if the test logic works properly at
count.

Test:

When the digital clock was logic gates and on
B.C. was changed and simulated and a
display for the digital clock was
shown. The counter was shown in hours, minutes and
seconds. The counter was shown in hours, minutes and
seconds. The counter was shown in hours, minutes and
seconds.